Hasnat Rafi Uddin

01863099966 — hasnatrafiuddinmunif@gmail.com
Bangladesh — Open to Remote
linkedin — github — portfolio — leetcode

Technical Skills

Languages: Python, JavaScript, SQL
QA & Testing: Test case design, edge case analysis, manual testing, automation scripting, API testing (REST)
Web & Tools: HTML5, CSS3, React, Git, Next.js, Stripe, REST APIs, Postman (basic), MongoDB (basic)
Machine Learning & AI: PyTorch, TensorFlow, YOLOv8, MediaPipe, OpenCV, CUDA
Concepts: AI/ML model validation, response accuracy analysis, logical error detection, system robustness

Education

B.Sc. in Computer Science — 2021 – 2025
BRAC University — *Bangladesh*

Higher Secondary Certificate (HSC) — 2020
BCIC College — *Bangladesh*

Secondary School Certificate (SSC) — 2018
Noakhali Zilla School — *Bangladesh*

Projects

ShopHub — *Next.js, MongoDB, React, Stripe, Tailwind CSS* — GitHub

- Developed a production-ready e-commerce application using Next.js, React, MongoDB, and TailwindCSS with JWT authentication, shopping cart, wishlist, and order management. Built a secure Admin CMS dashboard with product CRUD, user management, order tracking, and Cloudinary image uploads using RESTful APIs and Mongoose.

Movie Mania — *React, Vite, TMDB API* — GitHub

- Built a responsive movie browsing application using React, Vite, and React Router with live search, genre filtering, and trending/top-rated movie discovery via the TMDB API. Implemented favorites management using localStorage, dynamic carousel integration (Swiper), and optimized API fetching for performance.

PDF Reader with Hand Gesture Control — *JavaScript, PDF.js, MediaPipe, TensorFlow* — GitHub

- A web-based PDF reader that enables touchless navigation using real-time hand gesture recognition. It uses PDF.js for rendering and MediaPipe's GestureRecognizer to detect gestures like "One" (next page) and "Two" (previous page) via webcam.

Helmet & License Plate Detection — *Python, PyTorch, YOLOv8* — GitHub

- A deep learning-based system that detects helmet usage and recognizes license plate numbers using computer vision techniques on a custom dataset.

Undergraduate Thesis

Mitigating Domain Shift in Skin Cancer Recognition with Dropout Consistent Federated Averaging

- Reproduced three state-of-the-art skin lesion classification models.
- Designed an EfficientNet-based architecture with SE attention.
- Converted centralized pipelines to federated learning.
- Resolved dropout-induced zero dilution in FedAvg.
- Implemented mask-aware aggregation to improve robustness.